

ZP-DOC-03

Software Requirements Specification

Functional, non-functional and business-rule requirements for ZedPath Release 1 and 2

Version	1.0
Status	Approved
Date	5 October 2026
Classification	Public
Document owner	B.M.P Banneka, Product Owner

© 2026 Team Kestrel. Published under the classification shown above.

Uncontrolled when printed or exported. The controlled version is the Markdown source in the repository: docs/03-requirements/ZP-DOC-03_Software-Requirements-Specification.md

Document control

Document information

Document ID	ZP-DOC-03
Title	Software Requirements Specification
Version	1.0
Status	Approved
Classification	Public
Document owner	B.M.P Banneka, Product Owner
Author	Team Kestrel (drafted with Claude, AI coding partner)
Conforms to	ISO/IEC/IEEE 29148:2018, software requirements specification content; ZP-DOC-00
Source file	docs/03-requirements/ZP-DOC-03_Software-Requirements-Specification.md
Repository	https://github.com/PraveenBanneka/zedpath

Revision history

Version	Date	Author	Description of change
0.9	5 Oct 2026	Team Kestrel	First issue: functional requirements derived from ZP-DOC-02; NFRs; business rules from the UGC handbook and cut-off table 2025/26; submitted for approval
1.0	5 Oct 2026	B.M.P Banneka	Approved by the product owner; baselined

Review and approval

This document takes effect when the approver signs below. Until then its status is as shown on the cover.

Role	Name	Signature	Date
Reviewer	Claude (AI coding partner), consistency and traceability review		
Approver	B.M.P Banneka, Product Owner	Approved in writing (project channel)	5 Oct 2026

Contents

1	Introduction	6
1.1	Purpose	6
1.2	Scope.....	6
1.3	Intended audience	6
1.4	Relationship to other documents	6
1.5	Conventions	6
2	Overall description.....	7
2.1	Product perspective	7
2.2	Product functions.....	7
2.3	User classes and characteristics.....	7
2.4	Operating environment	7
2.5	Design and implementation constraints.....	7
2.6	Assumptions and dependencies	8
3	External interface requirements	9
3.1	User interfaces	9
3.2	Software interfaces	9
3.3	Communications interfaces	9
4	Functional requirements.....	10
4.1	FE-1 Profile and results	10
4.2	FE-2 State-university course discovery	10
4.3	FE-3 Preference list	11
4.4	FE-4 Journey and deadlines	12
4.5	FE-5 Other routes.....	12
4.6	FE-6 Ask.....	13
4.7	FE-7 People around the student	13
4.8	FE-8 After selection	13
4.9	FE-9 Data and trust	14
5	Non-functional requirements	15
5.1	Performance	15
5.2	Capacity and cost	15
5.3	Usability, accessibility and localisation	15
5.4	Security and privacy.....	16
5.5	Reliability and availability	16
5.6	Data quality.....	16
5.7	Maintainability and portability	16

6	Business rules.....	17
6.1	General eligibility	17
6.2	Streams, subjects and grades	17
6.3	Selection and the preference list	18
6.4	Course-specific requirements	18
6.5	Cut-off data	19
6.6	Banding (ZedPath policy)	19
6.7	Preference-list checks (ZedPath policy, derived from BR-016).....	20
6.8	Known gaps in the official sources.....	20
7	Verification.....	21
	Appendix A: Traceability: stories to requirements	22
	Appendix B: Glossary	24
	Appendix C: References.....	25

List of tables

Table 1:	Position of this document in the ZedPath document set	6
Table 2:	User classes.....	7
Table 3:	Software interfaces.....	9
Table 4:	Functional requirements, FE-1 Profile and results.....	10
Table 5:	Functional requirements, FE-2 State-university course discovery	10
Table 6:	Functional requirements, FE-3 Preference list	11
Table 7:	Functional requirements, FE-4 Journey and deadlines.....	12
Table 8:	Functional requirements, FE-5 Other routes	12
Table 9:	Functional requirements, FE-6 Ask	13
Table 10:	Functional requirements, FE-7 People around the student	13
Table 11:	Functional requirements, FE-8 After selection (Release R2)	13
Table 12:	Functional requirements, FE-9 Data and trust.....	14
Table 13:	Performance requirements.....	15
Table 14:	Capacity and cost requirements	15
Table 15:	Usability, accessibility and localisation requirements	15
Table 16:	Security and privacy requirements	16
Table 17:	Reliability and availability requirements	16
Table 18:	Data quality requirements	16
Table 19:	Maintainability and portability requirements	16
Table 20:	Business rules, general eligibility for state-university admission	17
Table 21:	Business rules, streams, subjects and grades	17
Table 22:	Business rules, quotas, selection and the preference list.....	18
Table 23:	Business rules, course-specific requirements.....	18
Table 24:	Business rules, cut-off data.....	19
Table 25:	Business rules, banding (ZedPath policy, configurable).....	19
Table 26:	Business rules, preference-list checks	20
Table 27:	Story-to-requirement traceability (every story in ZP-DOC-02 is covered).....	22
Table 28:	Terms and abbreviations used in this document.....	24
Table 29:	Sources referred to in this document	25

1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) states precisely what ZedPath **shall** do and how well it **shall** do it. Every requirement is uniquely identified, verifiable, and traced to the user story it comes from (ZP-DOC-02). Designers use it to build the system; testers use it to prove the system is correct.

1.2 Scope

The SRS covers Release R1 (Gate 3, 11 October 2026) and Release R2 (after the buildathon). Each requirement states its release. Requirements for R2 are included so the design does not have to be reworked later.

1.3 Intended audience

Developers, testers, the product owner, and buildathon judges assessing engineering quality.

1.4 Relationship to other documents

Table 1: Position of this document in the ZedPath document set

Document	Relationship
ZP-DOC-01 Vision and Scope	Grandparent. Business objectives BO-1 to BO-6 become quality requirements here
ZP-DOC-02 User Stories	Parent. Every functional requirement cites the story and acceptance criteria it refines
ZP-DOC-04 Use Case Model	Child. Use cases realise groups of functional requirements
ZP-DOC-05 Data Design	Child. Business rules and data requirements here become entities and constraints
ZP-DOC-06 Software Architecture	Child. Non-functional requirements here drive architecture decisions
ZP-DOC-08 Test Plan	Child. Every requirement here maps to at least one test case

1.5 Conventions

- Requirements use **shall** (mandatory), **should** (recommended) and **may** (optional), as defined in ZP-DOC-00.
- FR-enn**: functional requirement (*e* = feature/epic number). **NFR-nnn**: non-functional requirement. **BR-nnn**: business rule, a fact or policy of the domain that the system must respect.
- Priority** uses MoSCoW (M, S, C), inherited from the parent story unless stated.
- Verification method** (ISO/IEC/IEEE 29148): **T** test, **D** demonstration, **I** inspection, **A** analysis.

2 Overall description

2.1 Product perspective

ZedPath is a new, self-contained web application. It is not part of, and does not connect to, any UGC or government system: it **reads** their public documents and helps students act on them. The system context is shown in ZP-DOC-01. It runs entirely on the Cloudflare free plan (ZP-DOC-06).

2.2 Product functions

At the highest level, ZedPath shall let a student: enter their results once (FE-1); see every reachable state-university course with a likelihood band and reasons (FE-2); build and check a UGC preference list (FE-3); follow a dated journey with reminders (FE-4); see every other route (FE-5); ask cited questions in three languages (FE-6); share with family and get help from others (FE-7); and prepare for university (FE-8). Behind these, ZedPath shall keep every fact sourced, checked and correctable (FE-9).

2.3 User classes and characteristics

Table 2: User classes

Class	Persona	Frequency	Technical skill	Key needs
Student	P1, P2	Intense for 4 to 8 weeks after results, then occasional	Smartphone-native; limited data	Accuracy, speed, own language
Family member	P3	Occasional, through shared cards	Low	A simple summary in Sinhala or Tamil
Teacher or counsellor	P4	Weekly in application season	Moderate	Class overview
Verified senior	P5	Occasional	High	Safe, verified space
Curator	P6	Daily in season, weekly otherwise	High	Fast review of extracted facts

2.4 Operating environment

- Client: mobile web browsers (Chrome for Android 120+, Samsung Internet 24+, Safari for iOS 17+) and current desktop browsers. Screen widths from 360 px.
- Server: Cloudflare Workers runtime and Cloudflare data services (free plan).

2.5 Design and implementation constraints

These constraints are binding. Their detail is in ZP-DOC-06.

- **CON-1** The system shall run within the Cloudflare **free plan** limits verified on 5 October 2026.
- **CON-2** The application, API and tooling shall be written in **TypeScript**.
- **CON-3** The user interface shall follow **Material Design 3**.
- **CON-4** The system shall not send email, SMS or WhatsApp messages (not available or not free).
- **CON-5** Official third-party documents shall be stored privately and cited, not redistributed.

2.6 Assumptions and dependencies

As stated in ZP-DOC-01 (AS-1 to AS-4, DE-1 to DE-3).

3 External interface requirements

3.1 User interfaces

- **UI-1** The interface shall be a responsive, mobile-first web application with bottom navigation of at most four destinations (Journey, Courses, Routes, Ask), as specified in ZP-DOC-07.
- **UI-2** Every screen shall offer a language switch between Sinhala, Tamil and English.
- **UI-3** Every displayed fact shall offer access to its source (document, page, date verified).

3.2 Software interfaces

Table 3: Software interfaces

ID	External system	Purpose	Protocol and format
SI-1	Google Gemini API, through Cloudflare AI Gateway	Read documents and photos; generate answers in three languages	HTTPS, JSON
SI-2	Cloudflare Workers AI	Embeddings and fallback inference	Workers binding
SI-3	Government Gazette website	Fetch weekly Gazette issues	HTTPS, PDF
SI-4	Web Push services of browsers	Deliver reminders	Web Push protocol (RFC 8030) with VAPID (RFC 8292)
SI-5	Device calendar	Import deadlines	iCalendar file (RFC 5545)
SI-6	Device share sheet	Share the family card	Web Share API
SI-7	Google Sign-In (R2)	Verify university accounts of seniors	OpenID Connect

3.3 Communications interfaces

- **CI-1** All traffic shall use HTTPS (TLS 1.2 or later).
- **CI-2** The public API shall exchange JSON (UTF-8).

4 Functional requirements

Each table lists the requirements of one feature. *Source* names the parent story and acceptance criteria.

4.1 FE-1 Profile and results

Table 4: Functional requirements, FE-1 Profile and results

ID	Requirement	Source	Pri	Ver
FR-101	The system shall let a student enter stream, district, Z-score and three subjects with grades, without an account.	US-101, AC-101.1	M	T
FR-102	The system shall reject a Z-score outside -4.0000 to $+4.0000$ or with more than four decimal places, with an inline message.	AC-101.2	M	T
FR-103	The system shall reject a subject combination that is not admissible for the selected stream (BR-010) and name the offending subject.	AC-101.3	M	T
FR-104	The system shall store the profile only on the student's device and restore it on the next visit.	AC-101.4	M	T
FR-105	The system shall recalculate every result immediately when any profile field changes.	AC-102.1	M	T
FR-106	The system shall delete the profile from the device on request.	AC-102.2	M	T
FR-107	The system shall estimate a Z-score range from expected marks, using at least the three most recent years of subject statistics, labelled <i>Estimate</i> .	US-103	S	T
FR-108	When an estimated range is used, bands shall be computed from the lower bound and marked as estimate-based.	AC-103.3	S	T
FR-109	The system shall read stream, subjects, grades, Z-score and district from a results-sheet photo and require the student to confirm each field before use.	US-104	S	D
FR-110	The system shall highlight every field read with confidence below 0.85 for the student to check.	AC-104.2	S	T
FR-111	The system shall delete an uploaded photo as soon as reading finishes, and record the deletion in the audit log.	AC-104.3	S	I
FR-112	The system shall switch all interface text between Sinhala, Tamil and English without losing the current screen or data.	US-105	M	D
FR-113	When an offering has an O/L requirement (BR-022), the system shall ask for the relevant O/L grades, evaluate them, and show the offering as <i>Check O/L requirement</i> until they are given.	US-201; BR-022	M	T

4.2 FE-2 State-university course discovery

Table 5: Functional requirements, FE-2 State-university course discovery

ID	Requirement	Source	Pri	Ver
FR-201	The system shall list every course offering for which the student meets the general eligibility rules (BR-001 to BR-004) and the course's specific subject and grade requirements (BR-020 to BR-024), excluding suspended offerings (BR-026).	US-201	M	T
FR-202	The eligibility check shall evaluate alternative requirements ("A or B") and combined requirements ("A and B") exactly as stated in the handbook.	AC-201.3	M	T

ID	Requirement	Source	Pri	Ver
FR-203	The eligibility check shall run on the server; the client may cache results but shall not be the only place the rules are applied.	US-201; DoD 2	M	I
FR-204	The system shall show the number of hidden offerings and, for each, the rule failed in plain language with a link to the handbook page.	US-202	M	T
FR-205	The system shall assign each eligible offering exactly one band (Safe, Likely, Reach, Out of range or Not enough data) using the banding rules BR-030 to BR-036.	US-203	M	T
FR-206	For each banded offering, the system shall show the signed difference between the student's Z-score and the latest cut-off (four decimals) and the trend (rising, steady, jumpy).	AC-203.3	M	T
FR-207	No screen or answer shall state or imply that admission is certain.	AC-203.4	M	I
FR-208	The system shall show, per offering, the cut-off for the student's district for each available intake year, with the student's Z-score as the last row.	US-204	M	T
FR-209	Every cut-off, rule, duration, medium and aptitude-test flag shown shall link to its source document and page and show the date verified.	AC-204.2	M	I
FR-210	The system shall filter offerings by band, university, field of study and distance from the student's district, and update the count.	US-205	S	T
FR-211	For a chosen dream offering, the system shall show the gap to its lowest cut-off in the available years and classify the gap as <i>small</i> (≤ 0.10) or <i>large</i> (> 0.10).	US-206	S	T
FR-212	For a dream offering, the system shall suggest up to three offerings in the same field that are Safe or Likely for the student.	AC-206.2	S	T
FR-213	The system shall compare two offerings side by side on band, duration, medium, campus, aptitude test and trend.	US-207	C	D
FR-214	The system shall show monthly living-cost estimates by campus with sample size and collection date.	US-208	C (R2)	D

4.3 FE-3 Preference list

Table 6: Functional requirements, FE-3 Preference list

ID	Requirement	Source	Pri	Ver
FR-301	The system shall let the student add eligible offerings to an ordered list and reorder them by drag and drop and by keyboard.	US-301	M	T
FR-302	The system shall refuse to add an offering for which the student is not eligible.	AC-301.3	M	T
FR-303	The system shall warn when a Safe offering is listed above a Likely or Reach offering (BR-040, BR-041).	US-302	M	T
FR-304	The system shall warn when the list contains no Safe offering (BR-042).	AC-302.2	M	T
FR-308	The system shall limit the list to 125 Uni-Codes with no duplicates (BR-043).	US-301; BR-017	M	T
FR-305	The system shall propose a corrected order that keeps the student's most-wanted offerings first and Safe ones last, applied only after the student confirms.	AC-302.3	M	T
FR-306	The system shall copy the list's uni-codes to the clipboard, one per line, in list order.	US-303	M	T

ID	Requirement	Source	Pri	Ver
FR-307	The system shall present the official application steps with the most common mistake for each, linked to the official instructions.	US-304	S	I

4.4 FE-4 Journey and deadlines

Table 7: Functional requirements, FE-4 Journey and deadlines

ID	Requirement	Source	Pri	Ver
FR-401	The system shall show the student's journey stages in date order, marking past stages done and placing the next open deadline at the top with days remaining.	US-401	M	T
FR-402	Every date shall be labelled <i>Confirmed</i> (with its source) or <i>Estimated</i> (with the basis, e.g. the previous year's date).	US-402	M	T
FR-403	When the list contains an offering with an aptitude or practical test (BR-025), the journey shall add the test's application and test dates.	US-403	M	T
FR-404	The system shall generate an iCalendar file of the journey's deadlines with alarms 7 days and 1 day before each.	AC-404.1	S	T
FR-405	The system shall send web-push reminders 7 days and 1 day before each deadline to students who opted in.	AC-404.2	S	T
FR-406	The system shall delete a push subscription when the student turns notifications off.	AC-404.3	S	T
FR-407	The system shall check special-intake categories against the student's answers and link each to its handbook section.	US-405	C (R2)	T
FR-408	The system shall let the student add a route's dates to the journey.	US-406	C (R2)	D

4.5 FE-5 Other routes

Table 8: Functional requirements, FE-5 Other routes

ID	Requirement	Source	Pri	Ver
FR-501	The system shall list non-UGC routes in seven groups (private and non-state degrees, higher national diplomas, professional qualifications, vocational courses, job examinations, study abroad, retry) and mark each as open or not open to the student with the reason.	US-501	M	T
FR-502	Each route shall show entry requirements, duration, approximate cost, intake timing and sources with verified dates.	US-502	M	I
FR-503	The system shall show newly announced job examinations whose education and age rules the student meets, with closing date and Gazette link.	US-503	M	T
FR-504	A job-examination notice extracted with low confidence shall not be shown until a curator approves it.	AC-503.2	M	T
FR-505	The system shall show the recognition status of a programme's awarding university and local approval, each with source and date checked.	US-504	S (R2)	I
FR-506	The system shall check the basic conditions of the government interest-free loan scheme.	US-505	S (R2)	T
FR-507	The system shall list government scholarships abroad open to school leavers with usual timing and requirements.	US-506	C (R2)	I

ID	Requirement	Source	Pri	Ver
FR-508	For a dream offering, the system shall show the Z-score gap, the student's attempt number out of the maximum (BR-004) and alternative routes.	US-507	S	T

4.6 FE-6 Ask

Table 9: Functional requirements, FE-6 Ask

ID	Requirement	Source	Pri	Ver
FR-601	The system shall answer a question in the language it was asked (Sinhala, Tamil or English).	US-601	M	T
FR-602	Answers about eligibility shall use the student's profile when the student has one.	AC-601.2	M	T
FR-603	Every factual answer shall cite at least one stored source (document and page).	US-602	M	T
FR-604	When no stored source supports an answer, the system shall say it does not know and point to an official contact, and shall not generate an unsupported answer.	AC-602.2	M	T
FR-605	The system shall decline to predict future cut-offs or guarantee admission, and offer the historical pattern instead.	US-603	S	T
FR-606	The system shall offer actions under an answer (add to list, add to journey, compare).	US-604	C (R2)	D

4.7 FE-7 People around the student

Table 10: Functional requirements, FE-7 People around the student

ID	Requirement	Source	Pri	Ver
FR-701	The system shall generate a share card (image and short text) in the chosen language and pass it to the device share sheet.	US-701	M	D
FR-702	The share card shall exclude the student's name and Z-score unless the student includes them.	AC-701.2	M	T
FR-703	The system shall show seniors' answers with course, year and a verified badge.	US-702	S (R2)	D
FR-704	The system shall verify a senior by a university Google account on a recognised domain, or by curator review of proof that is deleted after the decision.	US-703	S (R2)	T
FR-705	The system shall provide groups by batch and target course.	US-704	C (R2)	D
FR-706	The system shall show a teacher aggregate counts for students who shared with the teacher's class code, and names only of those who shared.	US-705	C (R2)	T

4.8 FE-8 After selection

Table 11: Functional requirements, FE-8 After selection (Release R2)

ID	Requirement	Source	Pri	Ver
FR-801	The system shall present each university's registration requirements as a checklist with where to obtain each item.	US-801	C	I
FR-802	The system shall explain Mahapola and bursary eligibility and the documents that delay applications.	US-802	C	I

ID	Requirement	Source	Pri	Ver
FR-803	The system shall propose a month-by-month plan for the period before lectures, based on the course and district.	US-803	C	D
FR-804	The system shall keep a documents checklist covering every route.	US-804	C	D

4.9 FE-9 Data and trust

Table 12: Functional requirements, FE-9 Data and trust

ID	Requirement	Source	Pri	Ver
FR-901	The system shall let a curator register a source document with publisher, title, edition, issue date and language, store it unchanged, and record its SHA-256 fingerprint.	US-901	M	T
FR-902	The system shall detect a duplicate source by fingerprint.	AC-901.2	M	T
FR-903	The system shall extract, from a handbook source, one candidate rule per course with a confidence score and source page.	US-902	M	T
FR-904	An independent second model shall mark each candidate rule <i>agree</i> or <i>disagree</i> .	AC-902.2	M	T
FR-905	A candidate rule shall be published automatically only when both models agree and both confidences are at least 0.90; otherwise it shall enter the review queue.	AC-902.3	M	T
FR-906	The review queue shall show the source text, the proposed structured fact, both verdicts and the page, and let the curator approve, correct or reject.	US-903	M	D
FR-907	Each review decision shall be recorded with reviewer, time and the change made.	AC-903.2	M	T
FR-908	The system shall fetch new Gazette issues weekly, extract exam name, eligibility, age limit and closing date with confidence, and log counts per run.	US-904	M	T
FR-909	A failed scheduled run shall be retried automatically and its failure shown to the curator.	AC-904.2	M	T
FR-910	The system shall let anyone report a mistake on any fact with an optional note, linked to that fact, and queue it for the curator.	US-905	M	T
FR-911	The system shall keep every previous version of a published fact.	US-906	S	T
FR-912	The system shall require a bot check before accepting a report, a reminder opt-in or a question.	US-907	S	T
FR-913	The system shall limit Ask requests per client per minute (NFR-034) and return a friendly message when exceeded.	AC-907.2	S	T

5 Non-functional requirements

5.1 Performance

Table 13: Performance requirements

ID	Requirement	Ver
NFR-001	The first screen shall be interactive within 3 seconds on a "Fast 3G" network profile on a mid-range Android phone.	T
NFR-002	Eligibility and banding results shall appear within 2 seconds of submitting a profile on a "Fast 3G" profile.	T
NFR-003	95% of API requests (excluding Ask) shall complete within 500 ms at the edge.	T
NFR-004	Ask answers shall begin streaming within 4 seconds in 95% of cases.	T
NFR-005	Each API request shall use less than 10 ms of CPU time (the free-plan limit).	A
NFR-006	The initial JavaScript download shall not exceed 200 KB compressed.	I

5.2 Capacity and cost

Table 14: Capacity and cost requirements

ID	Requirement	Ver
NFR-010	The system shall serve 10,000 students per admission cycle within the free plan (BO-1, BO-6).	A
NFR-011	Daily usage of each Cloudflare free-plan quota shall stay below 70% on a typical day and below 100% on results day, according to the budget in ZP-DOC-06.	A
NFR-012	Page navigation shall not consume Worker requests (static assets are served without invoking the Worker).	I

5.3 Usability, accessibility and localisation

Table 15: Usability, accessibility and localisation requirements

ID	Requirement	Ver
NFR-020	A first-time user shall reach a checked preference list in under 10 minutes without help (BO-3), measured with at least 5 participants.	T
NFR-021	The interface shall meet WCAG 2.2 level AA.	I
NFR-022	Touch targets shall be at least 48 × 48 dp, following Material Design 3.	I
NFR-023	All interface text shall exist in Sinhala, Tamil and English; no screen shall mix languages except for official names and quotations.	I
NFR-024	Sinhala and Tamil text shall render correctly with Noto Sans Sinhala and Noto Sans Tamil.	D
NFR-025	Layouts shall work from 360 px width without horizontal scrolling.	T

5.4 Security and privacy

Table 16: Security and privacy requirements

ID	Requirement	Ver
NFR-030	The system shall store no personal data on the server without the student's explicit opt-in, in line with the Personal Data Protection Act No. 9 of 2022 (Sri Lanka).	I
NFR-031	Results photos shall be deleted within 5 minutes of upload, whether or not reading succeeds.	T
NFR-032	Curator functions shall require authentication; every curator action shall be logged.	T
NFR-033	Secrets (API keys) shall be stored as Cloudflare secrets and never in source code or the repository.	I
NFR-034	Ask shall accept at most 10 questions per client per minute and 100 per client per day.	T
NFR-035	All input shall be validated on the server; database access shall use bound parameters only.	I

5.5 Reliability and availability

Table 17: Reliability and availability requirements

ID	Requirement	Ver
NFR-040	Core browsing (courses, bands, list) shall remain usable if the AI services are unavailable.	T
NFR-041	Scheduled jobs shall retry failed steps at least three times with back-off.	T
NFR-042	The database shall be restorable to any point in the previous 7 days (D1 Time Travel).	D

5.6 Data quality

Table 18: Data quality requirements

ID	Requirement	Ver
NFR-050	100% of published facts shall reference a source document and page (BO-2).	A
NFR-051	Cut-off values shall be stored and compared as exact four-decimal values, not binary floating point.	I
NFR-052	Imported cut-off tables shall be verified by two independent extraction methods with zero unexplained differences before publication.	T
NFR-053	A reported mistake shall be triaged within 48 hours during the admission season.	A

5.7 Maintainability and portability

Table 19: Maintainability and portability requirements

ID	Requirement	Ver
NFR-060	Business rules (eligibility, banding thresholds) shall be data or pure functions with unit tests, not scattered through the interface.	I
NFR-061	Rule-engine code shall have at least 90% branch coverage.	A
NFR-062	A new intake year's handbook and cut-offs shall be loadable without code changes.	D
NFR-063	The front end shall run in any standards-compliant browser without installation; it may be installed as a progressive web app.	D

6 Business rules

Business rules are facts and policies of the admission domain that the system must respect. Their source is the UGC handbook and cut-off table for the 2025/26 intake; rules marked *ZedPath policy* are product decisions.

Sources are cited as **HB p.n** (printed page of the UGC handbook 2025/26 [2]; PDF page = printed + 7) and **COP** (UGC cut-off table 2025/26 [3]). Every handbook rule below was checked word for word against the PDF text. Rules marked **ZedPath policy** are product decisions; they are configurable and reviewed each intake.

6.1 General eligibility

Table 20: Business rules, general eligibility for state-university admission

ID	Rule	Source
BR-001	A candidate must have at least an S grade in each of three approved subjects in one sitting .	HB p.9
BR-002	A candidate must have at least 30% in the Common General Paper (also called the Common General Test), in the current attempt or a previous one within the three allowed; without it the candidate cannot register even if selected.	HB p.9
BR-003	A candidate may use at most three attempts ; after the third, results of any attempt cannot be used. Attempts need not be consecutive.	HB p.9, p.13
BR-004	Some candidates are ineligible regardless of results: those already registered as internal students of a UGC institution, degree holders, certain diploma and college students, holders of a government-channelled foreign scholarship, and anyone who made a false declaration.	HB p.13 to p.15
BR-005	A Z-score is entered with exactly four decimal places and must lie between -4.0000 and +4.0000 . (Published cut-offs for 2025/26 range from -0.8505 to 2.7935 .)	ZedPath policy; COP
BR-006	The candidate's district is the district of the school attended for the most days, totalling at least one year, in the three years before the examination; otherwise it is the district of permanent residence.	HB p.10, p.11
BR-007	Sixteen districts are educationally disadvantaged: Nuwara Eliya, Hambantota, Jaffna, Kilinochchi, Mannar, Mullaitivu, Vavuniya, Trincomalee, Batticaloa, Ampara, Puttalam, Anuradhapura, Polonnaruwa, Badulla, Monaragala and Ratnapura.	HB p.9

6.2 Streams, subjects and grades

Table 21: Business rules, streams, subjects and grades

ID	Rule	Source
BR-010	Subjects belong to six streams: Arts, Commerce, Biological Science, Physical Science, Engineering Technology and Biosystems Technology. A subject combination is admissible only if it follows the rules of its stream (for Arts, the basket rules 01 to 04).	HB p.20, p.31 to p.34, p.77, p.79
BR-011	Grades are ordered A > B > C > S > F . The handbook names S (Ordinary Pass), C (Credit Pass) and B (Very Good Pass); the full order is the Department of Examinations' grading scale.	HB p.9, p.83; ZedPath policy for the order

ID	Rule	Source
BR-012	The handbook gives no numeric subject codes; ZedPath assigns its own subject identifiers and maps every handbook subject name to one of them.	ZedPath policy

6.3 Selection and the preference list

Table 22: Business rules, quotas, selection and the preference list

ID	Rule	Source
BR-013	Up to 40% of places in a course are filled on all-island merit, up to 55% by district quota in proportion to population, and the remaining 5% are shared by the 16 educationally disadvantaged districts.	HB p.8, p.9
BR-014	Ten Arts courses (Arts, Arts SP, Arts SAB, Communication Studies, Peace and Conflict Resolution, Islamic Studies, Arabic Language, TESL, Social Work, Arts-IT) are filled 100% on all-island merit ; their cut-off is the same in every district (marked * in the COP).	HB p.8, p.21; COP
BR-015	Each course at each institution has a Uni-Code : a three-digit course code plus one institution letter (for example, 016C). There are 255 Uni-Codes for 2025/26.	HB p.112
BR-016	The UGC selects a candidate for the highest-preferred Uni-Code whose places are not already filled by candidates with higher Z-scores; a Uni-Code the candidate did not list is never allocated.	HB p.112, p.113
BR-017	A candidate may list at most 125 Uni-Codes and may not list the same Uni-Code twice.	HB p.113
BR-018	A candidate who does not register for a selected Uni-Code is not considered for the same course at any other university in later rounds, even if preferred higher.	HB p.114
BR-019	The order may be changed once , online, within two weeks of the closing date; the change replaces the whole list.	HB p.116, p.155

6.4 Course-specific requirements

Table 23: Business rules, course-specific requirements

ID	Rule	Source
BR-020	Each Uni-Code has subject and grade requirements that follow one of twenty patterns (fixed trio; trio with a count of higher grades; two fixed subjects plus a third from a list; one of several trios; slot rules; a gate subject at a higher grade; <i>n</i> of a list at a higher grade; and others). ZP-DOC-05 defines a rule grammar that expresses all twenty.	HB Section 2.2, p.31 to p.110
BR-021	Alternatives ("A or B") and combinations ("A and B") in a requirement are evaluated exactly as written; a requirement such as "a C grade in Combined Mathematics or Physics" is met by either subject.	HB p.70 (example: Computer Science)
BR-022	Fifty-one Uni-Codes also require G.C.E. (O/L) results (for example, a Credit Pass in English or Mathematics); some accept an A/L pass instead. The course text prevails over the summary table where they differ.	HB p.149, p.150; course pages

ID	Rule	Source
BR-023	Some requirements are not academic (Nursing: height at least 4 feet 10 inches and no deformity). They are shown as additional requirements and are not evaluated.	HB p.57, p.150
BR-024	Where a stream's "you may also apply for" list omits a course whose subject rule the student meets, the subject rule prevails .	ZedPath policy (handbook lists are incomplete)
BR-025	Twenty-four Uni-Codes require a practical or aptitude test run by the university; the student applies to the university directly after its press notice; passing does not remove the cut-off; failing affects only that course.	HB p.38, p.151
BR-026	Eight courses of the Gampaha Wickramarachchi University of Indigenous Medicine are suspended for 2025/26 and shall not be offered.	HB p.48, p.66, p.109
BR-027	Some courses are not taught in all three languages, so a student may meet the cut-off and still not be selected; this caveat is shown on every course page.	COP note; HB p.174

6.5 Cut-off data

Table 24: Business rules, cut-off data

ID	Rule	Source
BR-028	A cut-off is the minimum Z-score selected for one Uni-Code from one district in one intake year, published by the UGC after re-scrutiny. Previous cut-offs are guidance only and are presented as such.	HB p.10, p.112, p.173; COP
BR-029	NQC ("No Qualified Candidates") means no candidate from that district was selected for that course, for one of three stated reasons; it is not a numeric cut-off.	COP legend

6.6 Banding (ZedPath policy)

Bands turn cut-off history into an honest likelihood. They never promise admission (FR-207). Let Z be the student's Z-score (or the lower bound of an estimate), C the set of numeric cut-offs for the offering and the student's district over the most recent five intake years, n the number of values in C , and c_i the most recent value in C .

Table 25: Business rules, banding (ZedPath policy, configurable)

ID	Rule
BR-030	If $n = 0$ (every year NQC, or a new course), the band is Not enough data .
BR-031	If $n \geq 3$ the band uses <i>full history</i> ; if n is 1 or 2 it uses <i>limited history</i> and is labelled "based on n year(s)". The margin m is 0.0000 for full history and 0.1000 for limited history.
BR-032	Safe: $Z \geq \max(C) + m$. Likely: $c_i \leq Z < \max(C) + m$. Reach: $\min(C) - 0.0500 \leq Z < c_i$. Out of range: $Z < \min(C) - 0.0500$. Comparisons use exact four-decimal values (NFR-051).
BR-033	Trend (full history only): the least-squares slope of C by year; rising above +0.0200 per year, falling below -0.0200, otherwise steady ; jumpy if the standard deviation of C exceeds 0.1000 (overrides the others).
BR-034	An NQC year is excluded from C and shown as "No qualified candidates in <i>year</i> ".
BR-035	For all-island-merit offerings (BR-014) the single all-island cut-off is used for every district.

ID	Rule
BR-036	The thresholds 0.0500, 0.1000 and 0.0200 are configuration values. When three or more years are loaded they shall be validated by back-testing: bands computed from earlier years are compared with the following year's actual cut-offs.

6.7 Preference-list checks (ZedPath policy, derived from BR-016)

Table 26: Business rules, preference-list checks

ID	Rule
BR-040	Because selection takes the highest-listed Uni-Code the student clears (BR-016), an offering listed below one the student is very likely to clear is very unlikely to be reached.
BR-041	Ordering warning: when a Safe offering is listed above any Likely or Reach offering, the student is warned that the offerings below it are unlikely to be reached.
BR-042	Coverage warning: when the list contains no Safe offering, the student is warned that they may not be placed anywhere.
BR-043	The list enforces BR-017: at most 125 entries and no duplicates.

6.8 Known gaps in the official sources

The 2025/26 handbook does not state: application opening and closing dates (announced by press notice), test dates for aptitude tests, a tie-break rule, a re-scrutiny procedure, Mahapola or bursary eligibility rules, hostel allocation rules, or the registration deadline. ZedPath shows these as *Estimated* or *Not yet announced* (FR-402), never as confirmed facts. Subject means and standard deviations needed for the Z-score estimate (FR-107) are not in either source; FR-107 is therefore at risk until those statistics are obtained.

7 Verification

Each requirement states its verification method. ZP-DOC-08 maps every requirement to test cases and records results. A requirement is verified when its test cases pass, or when the inspection, analysis or demonstration is recorded with its evidence.

Appendix A: Traceability: stories to requirements

Table 27: Story-to-requirement traceability (every story in ZP-DOC-02 is covered)

Story	Title	Requirements
US-101	Enter my results once	FR-101, FR-102, FR-103, FR-104
US-102	Change my profile at any time	FR-105, FR-106
US-103	Estimate my Z-score before results	FR-107, FR-108
US-104	Read my results sheet from a photo	FR-109, FR-110, FR-111
US-105	Use the app in my language	FR-112
US-201	See every course I am eligible for	FR-113, FR-201, FR-202, FR-203
US-202	See why a course is hidden from me	FR-204
US-203	See Safe / Likely / Reach for each course	FR-205, FR-206, FR-207
US-204	Open a course page with sources	FR-208, FR-209
US-205	Filter and sort courses	FR-210
US-206	Work backwards from a dream course	FR-211, FR-212
US-207	Compare two courses	FR-213
US-208	See what living there costs	FR-214
US-301	Build my preference list	FR-301, FR-302, FR-308
US-302	Get warned about a bad order	FR-303, FR-304, FR-305
US-303	Copy uni-codes in order	FR-306
US-304	Learn the UGC form step by step	FR-307
US-401	See my journey with the next step first	FR-401
US-402	Know whether a date is confirmed or estimated	FR-402
US-403	Be told about aptitude tests for courses on my list	FR-403
US-404	Get reminders before deadlines	FR-404, FR-405, FR-406
US-405	Check whether a special intake applies to me	FR-407
US-406	Add my own steps to the journey	FR-408
US-501	See every route my results open	FR-501
US-502	Open a route's details	FR-502
US-503	Be alerted about job exams I qualify for	FR-503, FR-504
US-504	Check whether a degree is recognised	FR-505
US-505	Check the interest-free loan scheme	FR-506
US-506	See scholarships abroad	FR-507
US-507	Understand what retrying would take	FR-508
US-601	Ask a question in my language	FR-601, FR-602
US-602	Every answer cites its source	FR-603, FR-604
US-603	The assistant won't predict the future	FR-605
US-604	Turn an answer into an action	FR-606
US-701	Share a summary with my family	FR-701, FR-702
US-702	Ask a verified senior	FR-703

Story	Title	Requirements
US-703	Become a verified senior	FR-704
US-704	Join a group for my batch and target course	FR-705
US-705	See my class's progress as a teacher	FR-706
US-801	Get my registration checklist	FR-801
US-802	Check Mahapola and bursary	FR-802
US-803	Plan the gap year	FR-803
US-804	Keep my documents ready	FR-804
US-901	Register an official source	FR-901, FR-902
US-902	Extract rules from the handbook with AI	FR-903, FR-904, FR-905
US-903	Review uncertain facts	FR-906, FR-907
US-904	Scan the Gazette every week	FR-908, FR-909
US-905	Report a mistake	FR-910
US-906	Keep a history of every change	FR-911
US-907	Protect the app from abuse	FR-912, FR-913

Appendix B: Glossary

Table 28: Terms and abbreviations used in this document

Term	Meaning
Band	ZedPath's classification of an offering relative to the student's Z-score: Safe, Likely, Reach, Out of range, Not enough data
COP	The UGC table of minimum Z-scores ("cut-off points") by course, university and district
NQC	"No Qualified Candidates": a cut-off cell for which no candidate from that district was selected
Offering	One course of study at one university
SRS	Software Requirements Specification
VAPID	Voluntary Application Server Identification for Web Push (RFC 8292)

Appendix C: References

Table 29: Sources referred to in this document

Ref	Source
[1]	ISO/IEC/IEEE 29148:2018, <i>Requirements engineering</i>
[2]	University Grants Commission, <i>Admission to Undergraduate Courses of the Universities in Sri Lanka, Academic Year 2025/2026</i> (student handbook)
[3]	University Grants Commission, <i>Minimum "Z" Scores for selection to various Courses of Study, Academic Year 2025/2026</i> , 31 July 2026
[4]	Personal Data Protection Act, No. 9 of 2022, Democratic Socialist Republic of Sri Lanka
[5]	W3C, <i>Web Content Accessibility Guidelines (WCAG) 2.2</i>
[6]	IETF RFC 5545 (iCalendar), RFC 8030 (Web Push), RFC 8292 (VAPID)
[7]	ZP-DOC-02 User Stories; ZP-DOC-06 Software Architecture